

# List of Scientific Publications by Himanshu Kumar Dwivedi

[Google Scholar](#), [ORCID](#), [ResearchGate](#)

Required information: Publication(s) resulting from the doctoral dissertation: A1,A3,A4,A5,A6,A7,

\* = corresponding author;

## A) Peer-Reviewed: published or accepted for publishing

1. Kumar, S. **Dwivedi, H.K.\***, Rajeev (2026). "A deep learning WKB-PINNs framework for forward and inverse problems of semiclassical nonlinear Schrödinger equations. *Commun. Nonlinear Sci. Numer. Simul.*(Accepted)
2. Kumar, S. **Dwivedi, H.K.\***, Rajeev (2026). " Accelerated singularity-aware fractional PINNs for time-fractional phase-field equations: forward and inverse problems. *J. Comput. Sci.*, 100, 102969. <https://doi.org/10.1016/j.jocs.2026.102969>
3. Kumar, S. **Dwivedi, H.K.\***, Rajeev(2026) A novel time-discrete physics-informed neural network framework for forward and inverse problems of the 2D tempered fractional generalized Burgers equation. *Comput. Math. Appl.* 219, 85-109. <https://doi.org/10.1016/j.camwa.2026.07.003>
4. Kumar, S. **Dwivedi, H.K.**, M. Ehrhardt, Rajeev(2026) A Fractional-Memory Physics-Informed Neural Network with Fast History Compression for Tempered Fractional Coupled Phase-Field Systems. arXiv:2606.22191v1, <https://doi.org/10.48550/arXiv.2606.22191>
5. **Dwivedi H.K.\***. and Rajeev, (2026)An adaptive finite difference/ ADI scheme with maximum principle preservation for 2D Allen-Cahn equation with accelerated Caputo derivative evaluation, *Comput. Math. Appl.* 217, 402-429. <https://doi.org/10.1016/j.camwa.2026.06.018>
6. **Dwivedi, H.K.**, M. Ehrhardt, Rajeev, (2026)Alikhanov-XfPINNs: Adaptive Physics-Informed Learning for Nonlinear Fractional PDEs on Nonuniform Meshes. arXiv:2605.01305v1: <https://doi.org/10.48550/arXiv.2605.01305>
7. **Dwivedi, H.K.**, Rajeev, Zeng S. (2026). Nonuniform tempered Alikhanov scheme for fractional Allen-Cahn equations with discrete maximum principle. *Journal of Scientific Computing*, 107(1), 17. <https://doi.org/10.1007/s10915-026-03225-8>
8. Su, G., **Dwivedi, H.K.**,Pan, J. (2026). An accelerated Alikhanov-DeepONet operator learning framework for time-fractional phase field models. *Journal of Computational and Applied Mathematics*, 117711. <https://doi.org/10.1016/j.cam.2026.117711>
9. Kumar, S., **Dwivedi, H.K.\*** , Rajeev. (2026). Structure preserving conservative PINNs for semiclassical Schrodinger-type equations. *Journal of Computational and Applied Mathematics*, 485,117607. <https://doi.org/10.1016/j.cam.2026.117607>
10. **Dwivedi , H.K.**, Peng, Y., Rajeev, Zeng, S. (2026). Finite difference/deep learning scheme for tempered fractional generalized Burgers' equation with fast evaluation of Caputo derivative. *Applied Mathematics and Computation*, 523, 130032. <https://doi.org/10.1016/j.amc.2026.130032>
11. **Dwivedi, H.K.**, Rajeev, Zeng, S.(2026). An alternating direction implicit method for 2D nonlinear Schrödinger equation with accelerated evaluation of Caputo derivative. *Applied Mathematics and Computation*, 518, 129905. <https://doi.org/10.1016/j.amc.2025.129905>
12. **Dwivedi, H.K.\***,Rajeev. (2025). A novel accelerated tempered algorithm with nonuniform time-stepping compact ADI schemes for 2D tempered-fractional nonlinear Schrödinger equations with weak initial

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singularity. *Computers and Mathematics with Applications*, 196,312-337. <https://doi.org/10.1016/j.camwa.2025.07.036>

13. **Dwivedi, H.K.**, Rajeev. (2025). A novel fast second order approach with high-order compact difference scheme and its analysis for the tempered fractional Burgers equation. *Mathematics and Computers in Simulation*, 227, 168–188. <https://doi.org/10.1016/j.matcom.2024.08.003>
14. **Dwivedi, H.K.**, Rajeev.(2024). A novel fast tempered algorithm with high-accuracy scheme for 2D tempered fractional reaction-advection-subdiffusion equation. *Computers and Mathematics with Applications*,176, 371–397. <https://doi.org/10.1016/j.camwa.2024.11.004>
15. **Dwivedi, H.K.\***, Rajeev.(2024). Fast high-order linearized exponential methods for efficient simulation of 2D time-fractional Burgers equation in polymer solution dynamics. *Journal of Mathematical Chemistry*, vol. 63, pp. 596–625, 2025. <https://doi.org/10.1007/s10910-024-01682-w>
16. **Dwivedi, H.K.**, Rajeev. (2024). A fast difference scheme for the multi-term time fractional advection–diffusion equation with a non-linear source term. *Chinese Journal of Physics*, 89, 86–103. <https://doi.org/10.1016/j.cjph.2024.02.051>
17. **Dwivedi H.K.**, Mistri, R.K., (2022). Direct and inverse problems for subset sums with certain restrictions. *Integers: Electronic Journal of Combinatorial Number Theory*, 22,1–13. DOI: 10.5281/zenodo.10999274

### B) Submitted manuscript with peer-review

1. **Dwivedi, H.K.\***, Rajeev, (2026). Finite difference/finite element scheme for time fractional dispersion equation with fast evaluation of Caputo derivative.
2. **Dwivedi, H.K.\***, Rajeev,(2026) A fast second order finite difference approach enhanced by physics informed neural network for solving the multi-dimensional tempered fractional Allen-Cahn equation.
3. **Dwivedi, H.K.\***, Kumar, S., Rajeev, (2026) An accelerated tempered finite difference approach enhanced by physics informed neural network for solving the multi-dimensional tempered fractional generalized Burgers equation.
4. **Dwivedi, H.K.,\*** Rajeev, (2026). A novel accelerated tempered Alikhanov formula with compact ADI and nonuniform time-stepping for tempered fractional nonlinear Schrödinger equations exhibiting weak initial singularity.
5. **Dwivedi , H.K.\***, Rajeev,(2026). A novel efficient compact ADI time-stepping scheme for nonuniform time-fractional Allen-Cahn equation exhibiting weak initial singularity.
6. **Dwivedi , H.K.\***, Rajeev, A.S. Hendy, A.A. Alikhanov (2026).Energy dissipative and maximum principle preserving accelerated numerical schemes for the tempered time fractional Allen--Cahn model.
7. **Dwivedi , H.K.\***, S. Kumar, Rajeev,(2026). Tempered-DeepONet: A Physics-Informed Asymptotic-Memory Operator Learning Framework for Weakly Singular Tempered Fractional PDEs.

### C) PhD Thesis

**Title: Analysis of Accelerated Finite-Difference and Scientific Machine Learning Schemes for Time-Fractional Partial Differential Equations. ([link](#))**